3.1.7 Oxidation, reduction and redox equations: Sheet 3

Combining half equations · 19 marks · about 25 minutes

Key information

- Write the two half equations, then multiply one or both so the number of electrons is the same in each.
- Add the half equations together and cancel the electrons; cancel any H^+ or H_2O that appear on both sides.
- Check: atoms and total charge balance in the overall equation, and there are no electrons left.
- The oxidising agent gains electrons (it is reduced); the reducing agent loses electrons (it is oxidised).

Section A: Self-quiz

01 Which statement about this redox reaction is correct?

$3\ Sn^{2+}(aq) + Cr_2O_7^{2-}(aq) + 2\ H^+(aq) \rightarrow 2\ Cr^{3+}(aq) + 3\ SnO_2(s) + H_2O(l)$

[1 mark]

A Sn^{2+} is the oxidising agent and it gains electrons.
B Sn^{2+} is the reducing agent and it gains electrons.
C $Cr_2O_7^{2-}$ is the oxidising agent and it gains electrons.
D $Cr_2O_7^{2-}$ is the reducing agent and it gains electrons.

02 In which oxide is the named element in its highest oxidation state?

[1 mark]

A chlorine in ClO_2
B magnesium in MgO
C nitrogen in N_2O_4
D sulfur in SO_2

03 What is the best oxidising agent?

[1 mark]

A F_2
B F^-
C I_2
D I^-

04 State, in terms of electrons, the meaning of the term oxidising agent.

[1 mark]

05 Iodide ions reduce copper(II) ions in aqueous solution to form a precipitate of copper(I) iodide (CuI) and iodine. Give an equation for this redox reaction.

[1 mark]

Section B: Exam questions

06 This question is about redox reactions.

06.1 Give a half-equation to show the oxidation of copper to copper(II) ions.

[1 mark]

06.2 Give a half-equation to show the reduction of NO_3^- ions in acidic solution to NO_2

[1 mark]

06.3 Use your answers to the two previous parts to deduce an overall equation for the reduction of NO_3^- ions by copper.

[1 mark]

07 $Cr_2O_7^{2-}$ can oxidise SO_3^{2-} in acidic conditions to form Cr^{3+} and SO_4^{2-}

Deduce a half-equation for the oxidation of SO_3^{2-} to SO_4^{2-}

Deduce a half-equation for the reduction of $Cr_2O_7^{2-}$ to Cr^{3+}

Deduce the overall equation for the oxidation of SO_3^{2-} by $Cr_2O_7^{2-}$

[3 marks]

08 In acidic conditions, ClO_3^- ions oxidise Cl^- ions to form Cl_2

Deduce a half-equation for the oxidation of Cl^- to Cl_2

Deduce a half-equation for the reduction of ClO_3^- to Cl_2

Deduce the overall equation for this reaction.

[3 marks]

09 When an acidified solution of sodium nitrite ($NaNO_2$) is added to aqueous potassium iodide, iodine and nitrogen monoxide (NO) are formed.

09.1 Write an overall ionic equation for the reaction of NO_2^- in an acidic solution with I^-

[1 mark]

09.2 State the role of NO_2^- in the reaction with I^-

[1 mark]

10 Solid sodium iodide reacts with concentrated sulfuric acid to form hydrogen sulfide.

Give a half-equation to show the oxidation of iodide ions.

Give a half-equation to show the reduction of concentrated sulfuric acid to hydrogen sulfide.

Use your half-equations to deduce an overall equation for this reaction.

[3 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**C**	Source: 7404/1 Jun 2023 Q17	1
02	**B**	Source: 7405/3 Jun 2023 Q14	1
03	**A**	Source: 7404/1 Jun 2019 Q19	1
04	Electron acceptor	Source: 7404/1 Nov 2021 Q03.1 Do not allow electron pair acceptor	1
05	$2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$	Source: 7405/1 Jun 2025 Q05.5 (products named)	1
06.1	$Cu \rightarrow Cu^{2+} + 2e^-$	Source: 7404/1 Jun 2024 Q09.2–09.4 Ignore state symbols. Accept multiples	1
06.2	$NO_3^- + 2H^+ + e^- \rightarrow NO_2 + H_2O$	Ignore state symbols. Accept multiples	1
06.3	$Cu + 2NO_3^- + 4H^+ \rightarrow Cu^{2+} + 2NO_2 + 2H_2O$	Ignore state symbols. Accept multiples	1
07	$SO_3^{2-} + H_2O \rightarrow SO_4^{2-} + 2H^+ + 2e^-$ $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ $3SO_3^{2-} + Cr_2O_7^{2-} + 8H^+ \rightarrow 3SO_4^{2-} + 2Cr^{3+} + 4H_2O$	Source: 7404/1 Nov 2021 Q03.2 Allow multiples in each case	3
08	M1 $2Cl^- \rightarrow Cl_2 + 2e^-$ M2 $2ClO_3^- + 12H^+ + 10e^- \rightarrow Cl_2 + 6H_2O$ M3 $2ClO_3^- + 12H^+ + 10Cl^- \rightarrow 6Cl_2 + 6H_2O$	Source: 7405/1 Jun 2024 Q05.3 Allow multiples M3 allow $ClO_3^- + 6H^+ + 5Cl^- \rightarrow 3Cl_2 + 3H_2O$	3
09.1	$2NO_2^- + 2I^- + 4H^+ \rightarrow I_2 + 2NO + 2H_2O$	Source: 7404/1 Jun 2017 Q08.4–08.5	1
09.2	Oxidising agent	Allow to accept / gain electrons; allow oxidant Do not allow accept / gain pairs of electrons; do not allow oxidise	1
10	Half-equation 1: $2I^- \rightarrow I_2 + 2e^-$ Half-equation 2: $H_2SO_4 + 8H^+ + 8e^- \rightarrow H_2S + 4H_2O$ OR $SO_4^{2-} + 10H^+ + 8e^- \rightarrow H_2S + 4H_2O$ Overall: $H_2SO_4 + 8H^+ + 8I^- \rightarrow 4I_2 + H_2S + 4H_2O$ OR $SO_4^{2-} + 10H^+ + 8I^- \rightarrow 4I_2 + H_2S + 4H_2O$ OR $H_2SO_4 + 8HI \rightarrow 4I_2 + H_2S + 4H_2O$	Source: 7404/1 Jun 2023 Q06.3 Ignore state symbols	3